Data Storytelling

Explaining data through narrative and interaction

Information Visualization · Lecture 13 · Week 7

Five parts of a data story

01	**Foundations**	Exploration and explanation
02	**Narrative sequence**	Three acts, four moves, and pacing
03	**Attention**	Emphasis, annotation, and reading paths within a view
04	**Balance**	Author guidance, reader exploration, structures and genres
05	**Storyboarding**	Planning scenes, transitions, and reader choices

Exploration and explanation serve different tasks

Data storytelling combines data visualizations with a narrative: an organized sequence of explanations that helps readers understand a topic or finding.

Exploration

The analyst varies views, asks new questions, and searches for patterns.

Typical experience: an open-ended analysis tool.

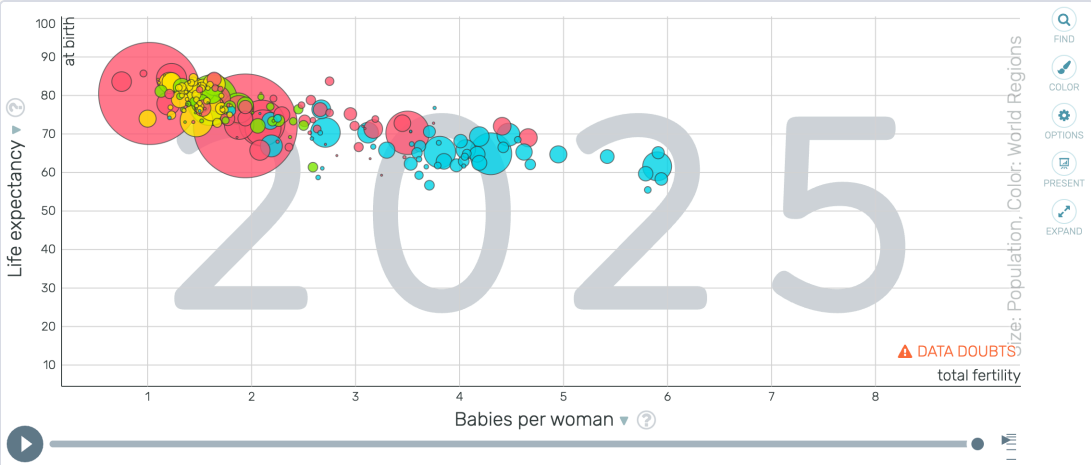
Explanation

The author arranges views and commentary to develop an understandable account.

Typical experience: a guided data story.

Interactive data stories place reader choices within an explanatory sequence. See Segel & Heer (2010), introduction.

A chart can be explored or narrated



Gapminder Tools · Open original ↗



Hans Rosling / BBC News · Watch the film ↗

Tools: the reader chooses variables and comparisons.

Film: narration and animation introduce the chart and move through time.

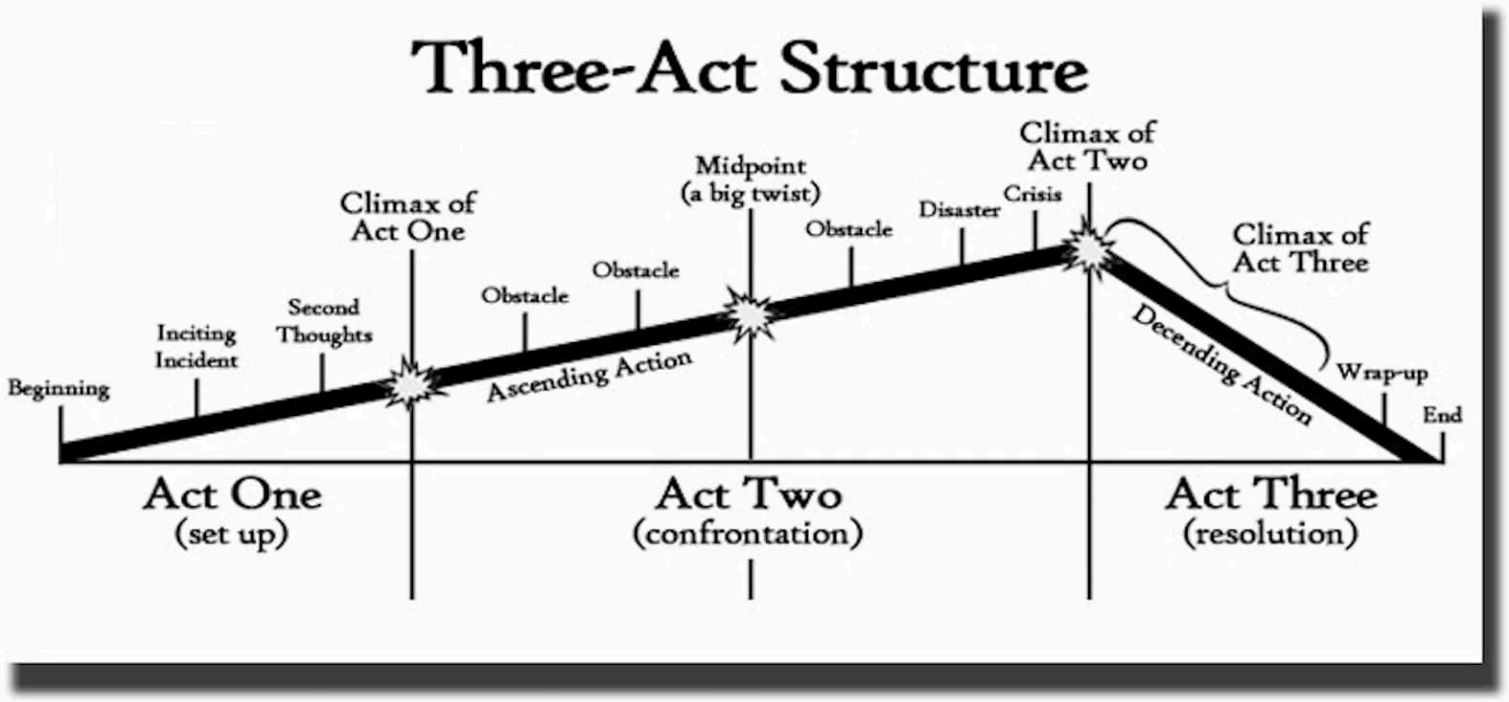
PART 02

Organizing a narrative sequence

Arrange information across views: build the explanation, control its pace, and preserve continuity.

Three acts establish, develop, and resolve

A **three-act narrative** divides a story into a setup, a development, and a resolution. Each act changes what the reader knows about the situation.



Act 1 · Setup

Introduce the setting and a question or tension.

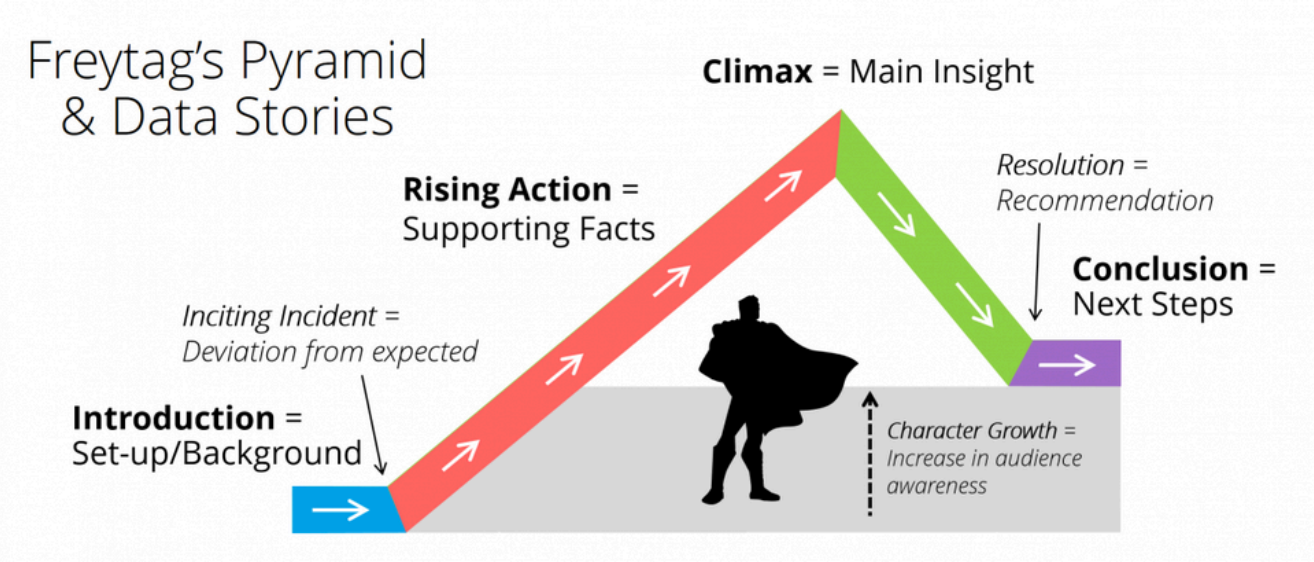
Act 2 · Development

Develop that tension through events and obstacles: the diagram's **confrontation**.

Act 3 · Resolution

Bring the story to an outcome and show what it means.

Data stories can unfold in four moves



Main Insight: the answer

The central answer that emerges from the evidence and resolves the story's question.

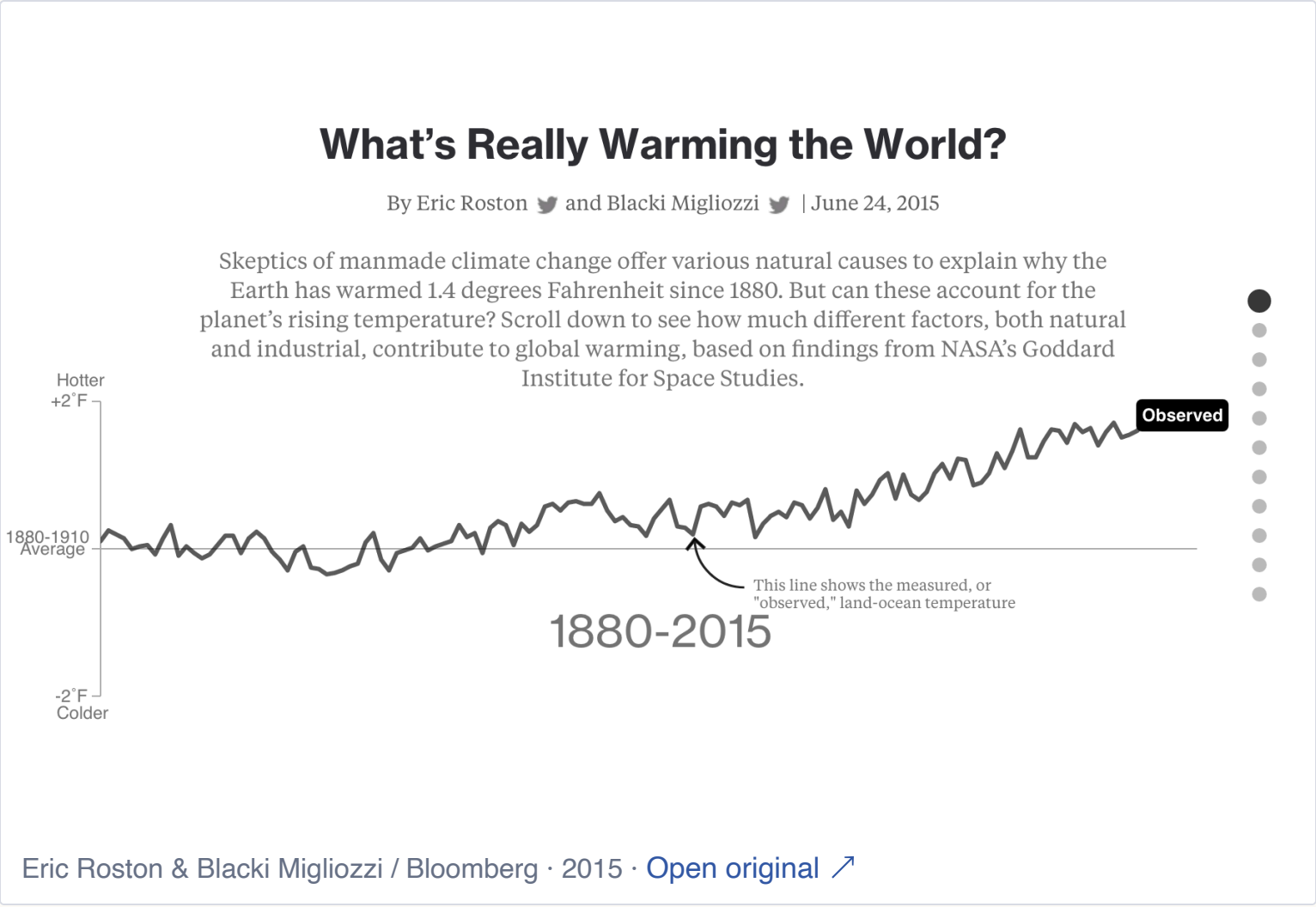
Implication: what it means

The answer's significance, consequences, or possible next steps.

1. Context	2. Question	3. Evidence	4. Implication
Introduction Set-up / Background	**Inciting Incident** Deviation from expected	**Rising Action → Climax** Supporting Facts → Main Insight	**Resolution + Conclusion** Recommendation / Next Steps

A teaching adaptation of the dramatic arc. An implication need not propose an action.

A sequence of questions about warming



Eric Roston & Blacki Migliozzi / Bloomberg · 2015 · Open original ↗

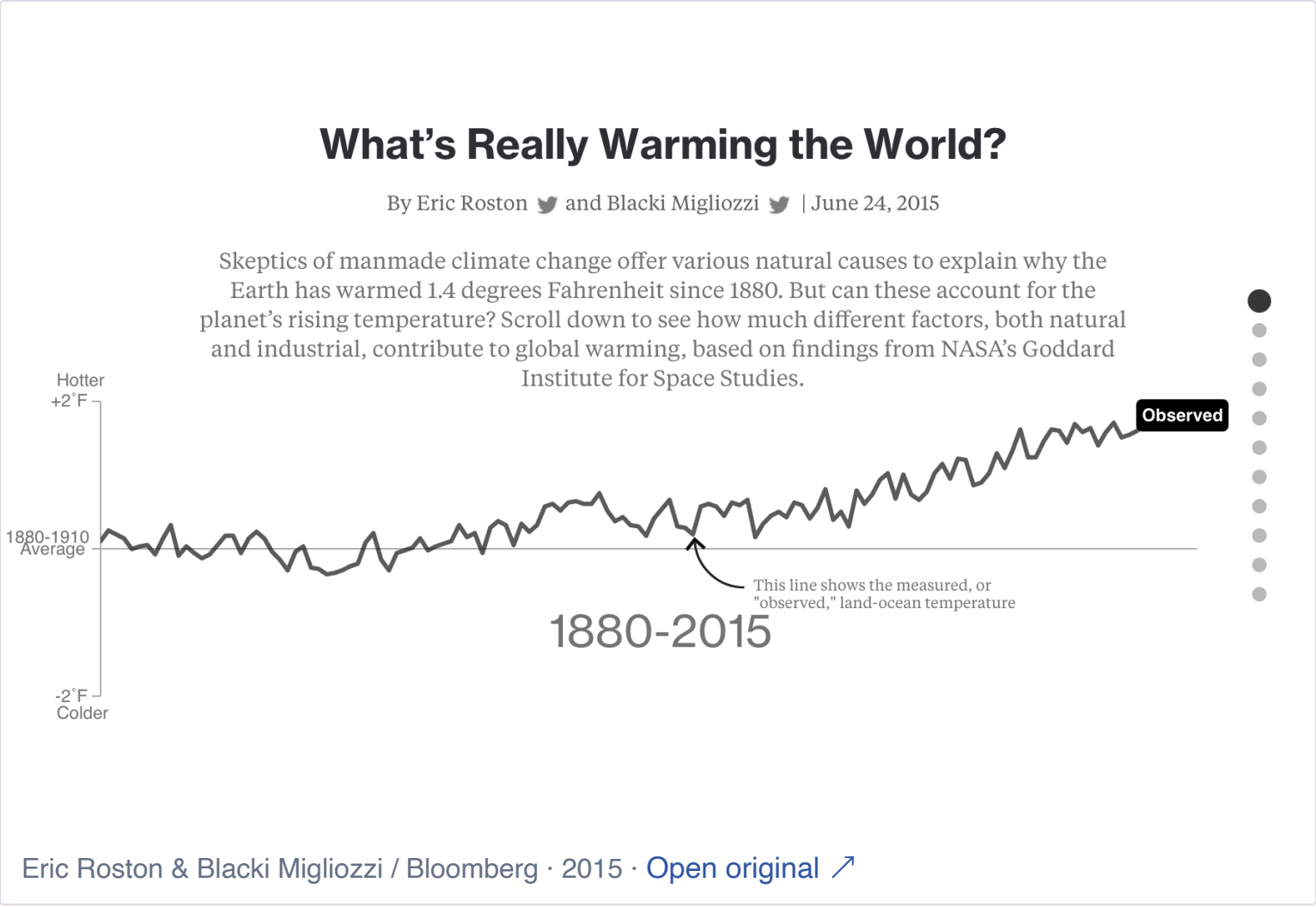
Follow the original

The opening introduces a temperature record and a question about its causes.

Later views consider different factors before bringing them together.

Watch the sequence: how do comparisons build toward an answer, and what does the ending ask next?

Context: start with the observed change



Eric Roston & Blacki Migliozzi / Bloomberg · 2015 · Open original ↗

01 · CONTEXT

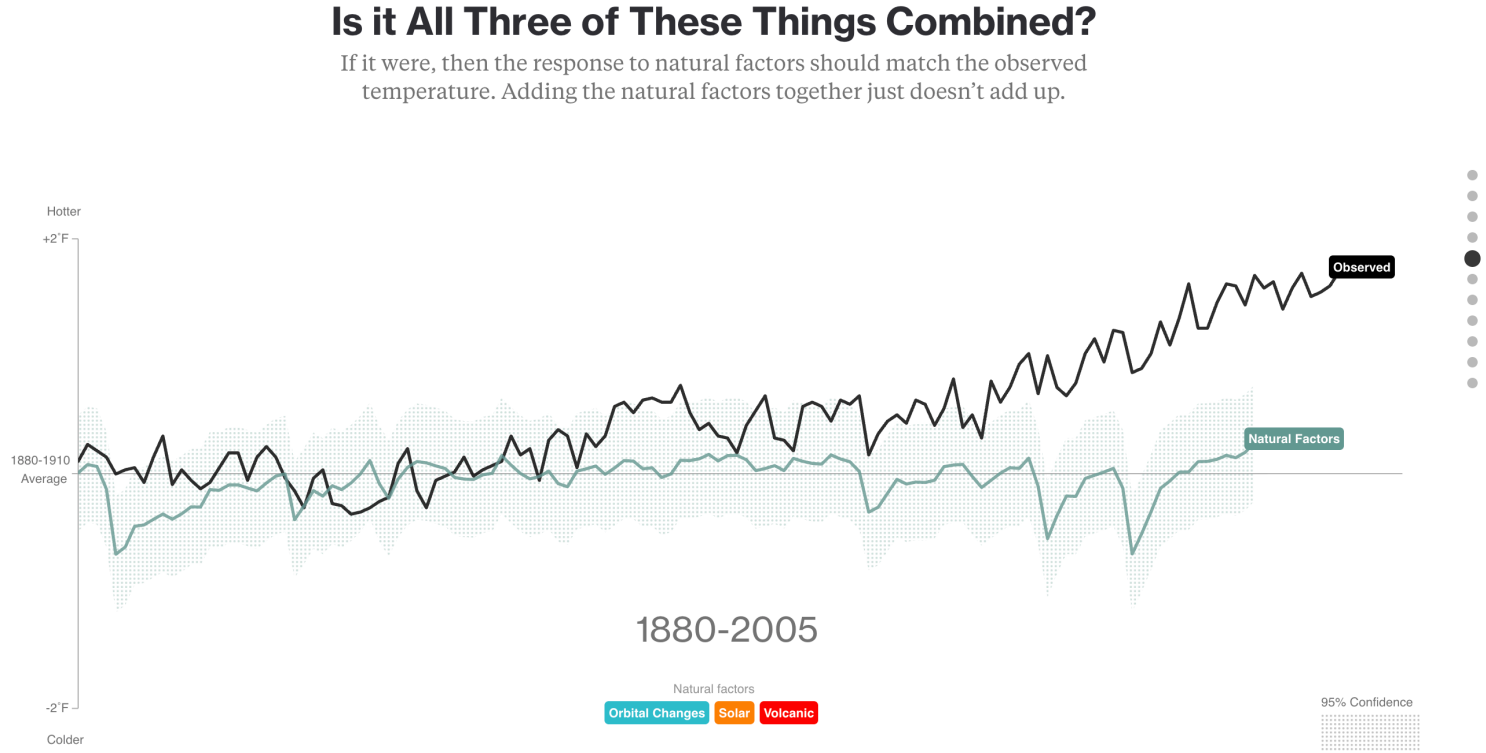
The observed record shows a long-term rise, especially toward the end of the period.

Readers meet the **observed temperature record** before any modeled factor appears.

This line becomes the reference for later views.

Question: could natural factors explain it?

Progressive disclosure introduces information in stages, making additional detail available as an explanation develops.



Eric Roston & Blacki Migliozzi / Bloomberg · 2015 · Open original ↗

02 · QUESTION

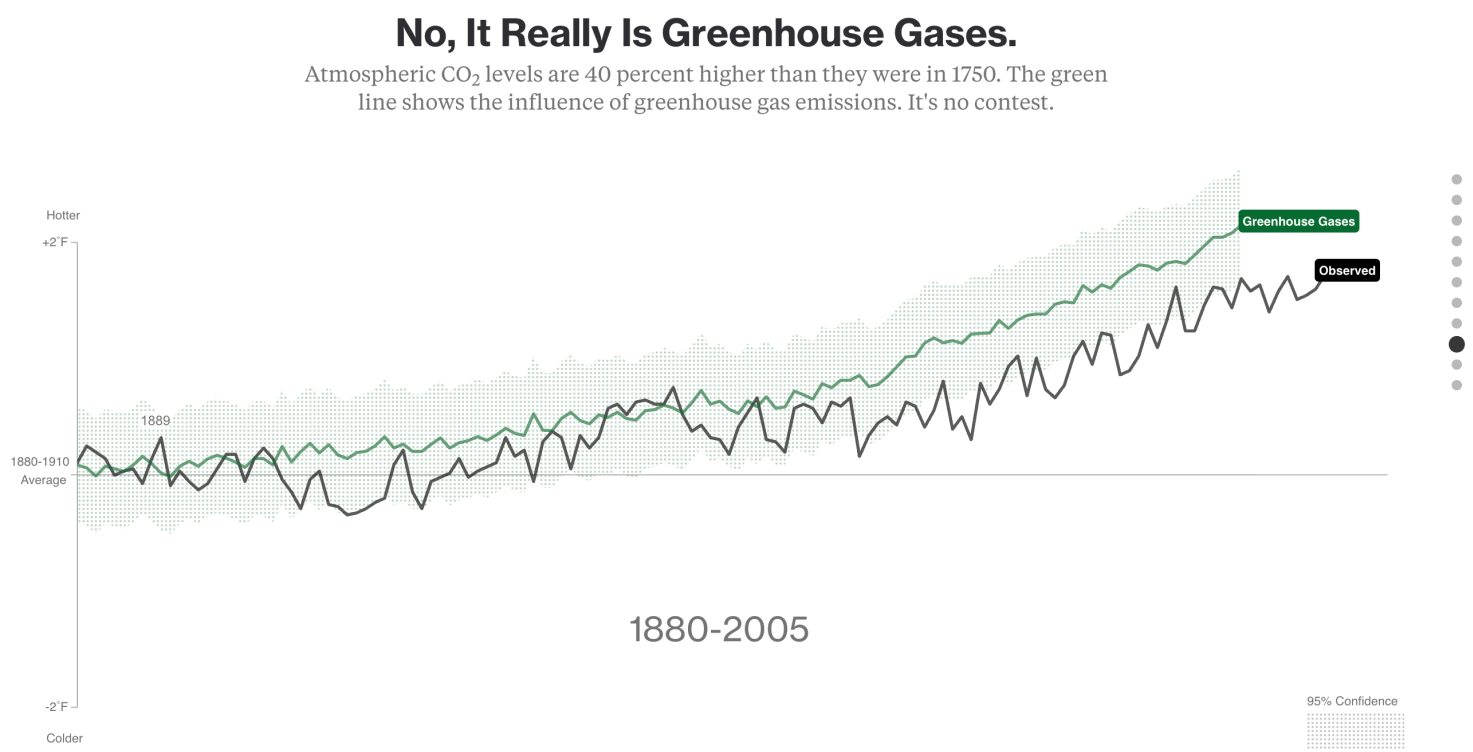
Natural factors enter first: orbit, solar variation, volcanoes, then their combined response.

Together, they do not reproduce the late-century rise.

Human factors enter in the following views, giving the explanation its next step.

Evidence: preserve continuity across comparisons

Object constancy is the perception that a visual object remains the same object as its appearance or surrounding view changes.



Eric Roston & Blacki Migliozzi / Bloomberg · 2015 · Open original ↗

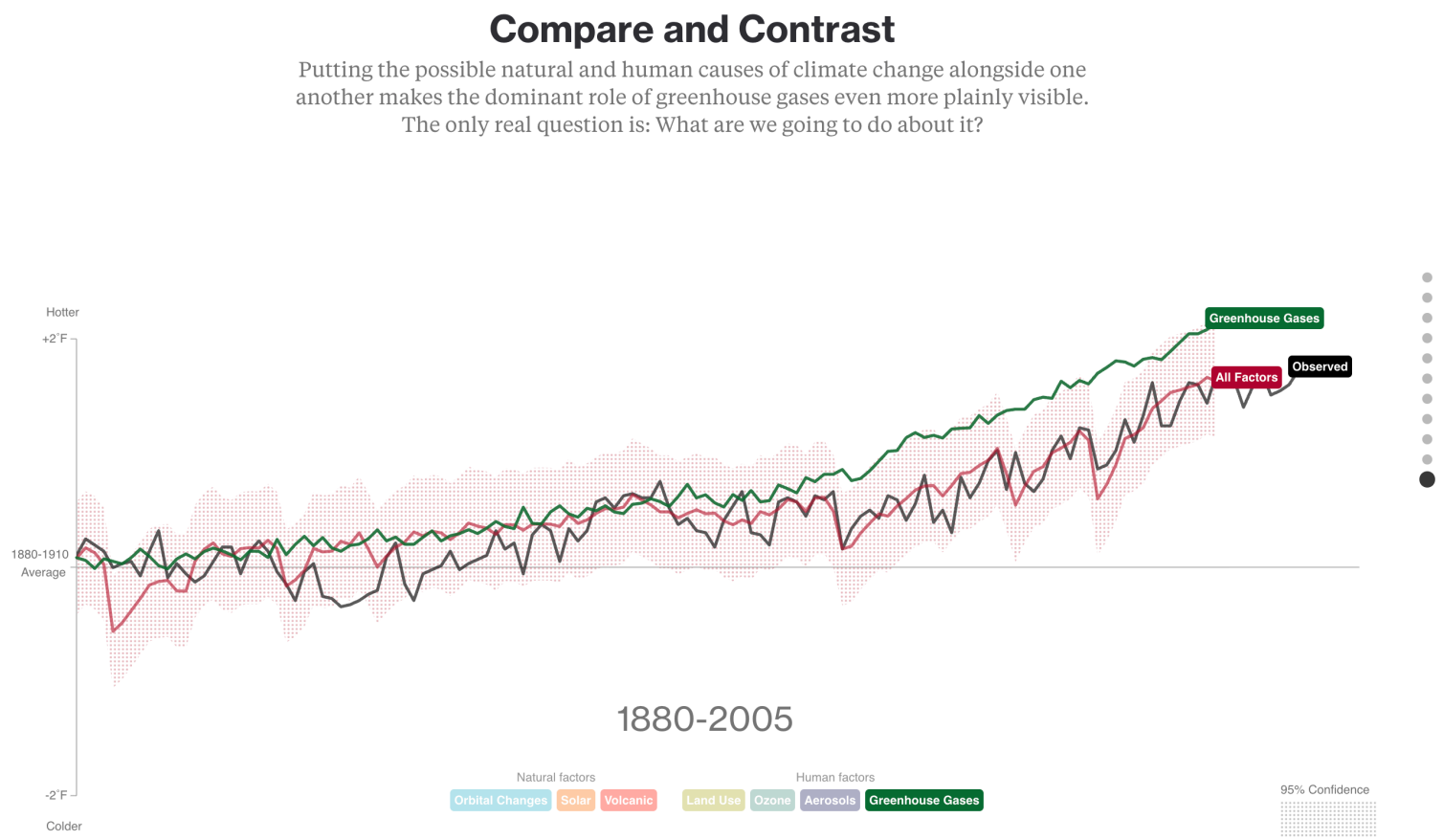
03 · EVIDENCE

The observed line and axes remain in place as the modeled factor changes.

Readers can carry the same reference into the greenhouse-gas comparison.

This new comparison builds toward the **main insight**.

Bloomberg: main insight and implication



Eric Roston & Blacki Migliozzi / Bloomberg · 2015 · Open original ↗

MAIN INSIGHT · THE ANSWER

Greenhouse gases are the dominant warming influence.

The combined model tracks the observed rise.

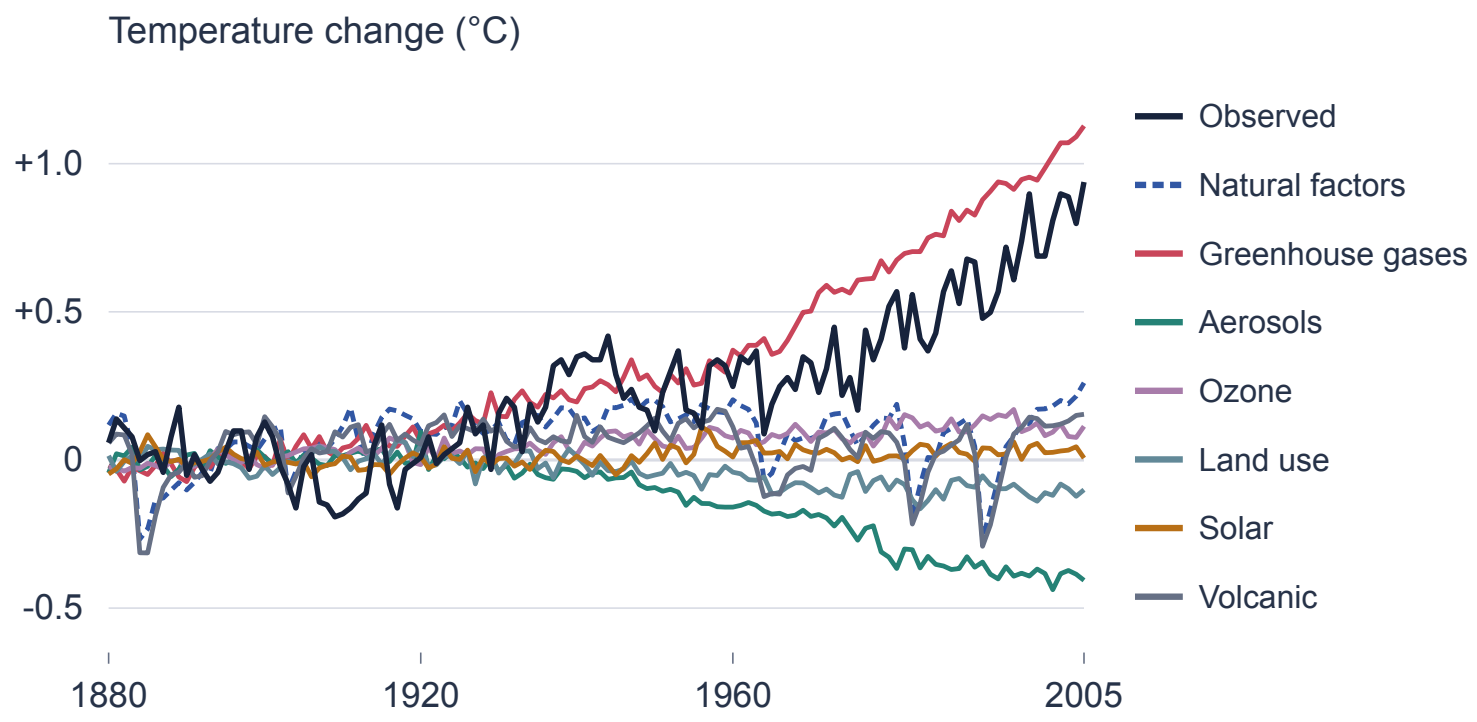
IMPLICATION · WHAT FOLLOWS

Attention turns to **how we respond**.

“What are we going to do about it?”

Bloomberg’s closing question

Pacing determines when each layer appears



Course replot · NASA GISS data distributed by Bloomberg · 1880–2005 · Open original ↗

Pacing is the timing and amount of information introduced at each stage of a story.

This view puts many factors on screen at once.

Progressive disclosure stages the comparisons; **object constancy** keeps the observed line as a familiar reference.

Compare: what would change if this were the first view?

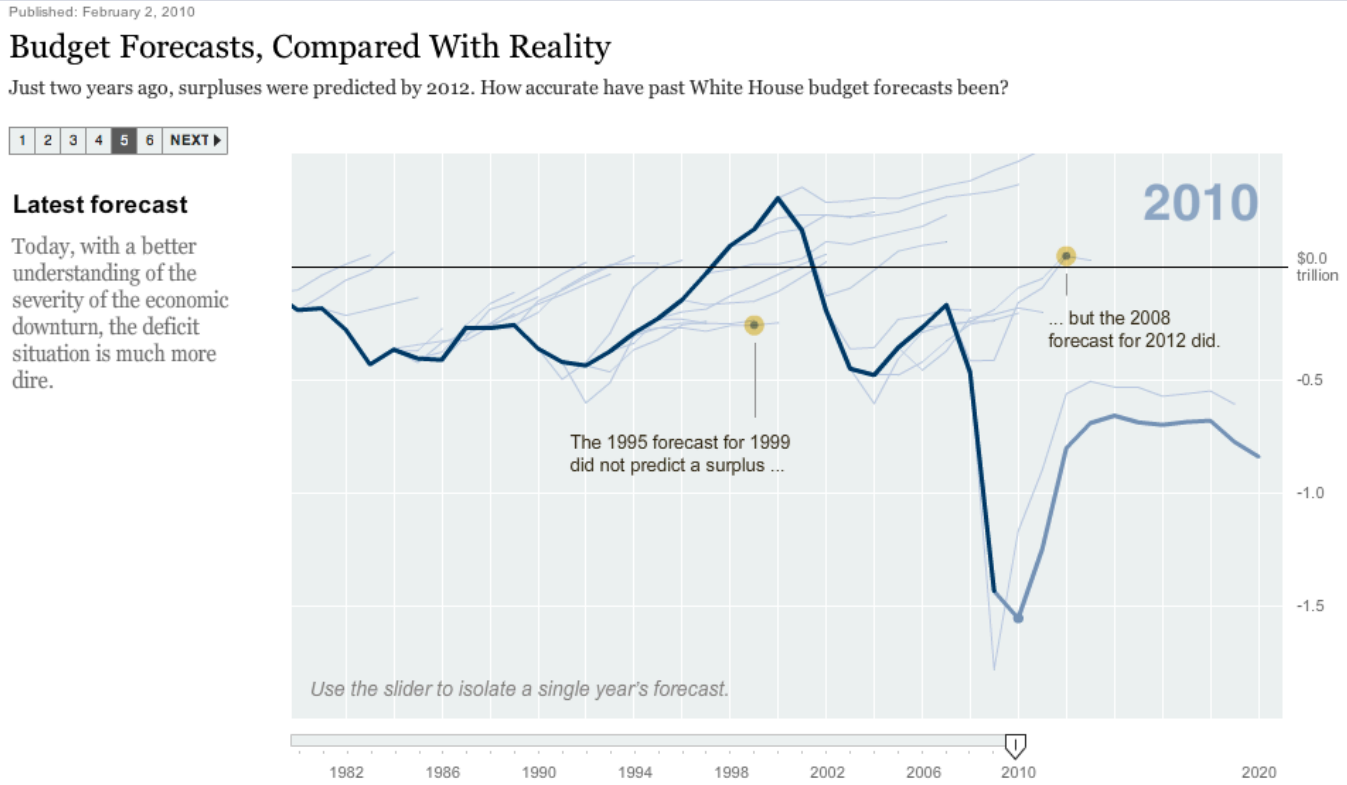
PART 03

Directing attention within a view

In a single static visualization, emphasis, annotations, and layout guide what readers notice and where they look next.

Visual emphasis makes a comparison stand out

Visual emphasis makes selected elements more noticeable through differences in color, contrast, size, or line weight, often by reducing the prominence of other elements.



The New York Times · 2010 · Figure 2 in Segel & Heer · Open archive ↗

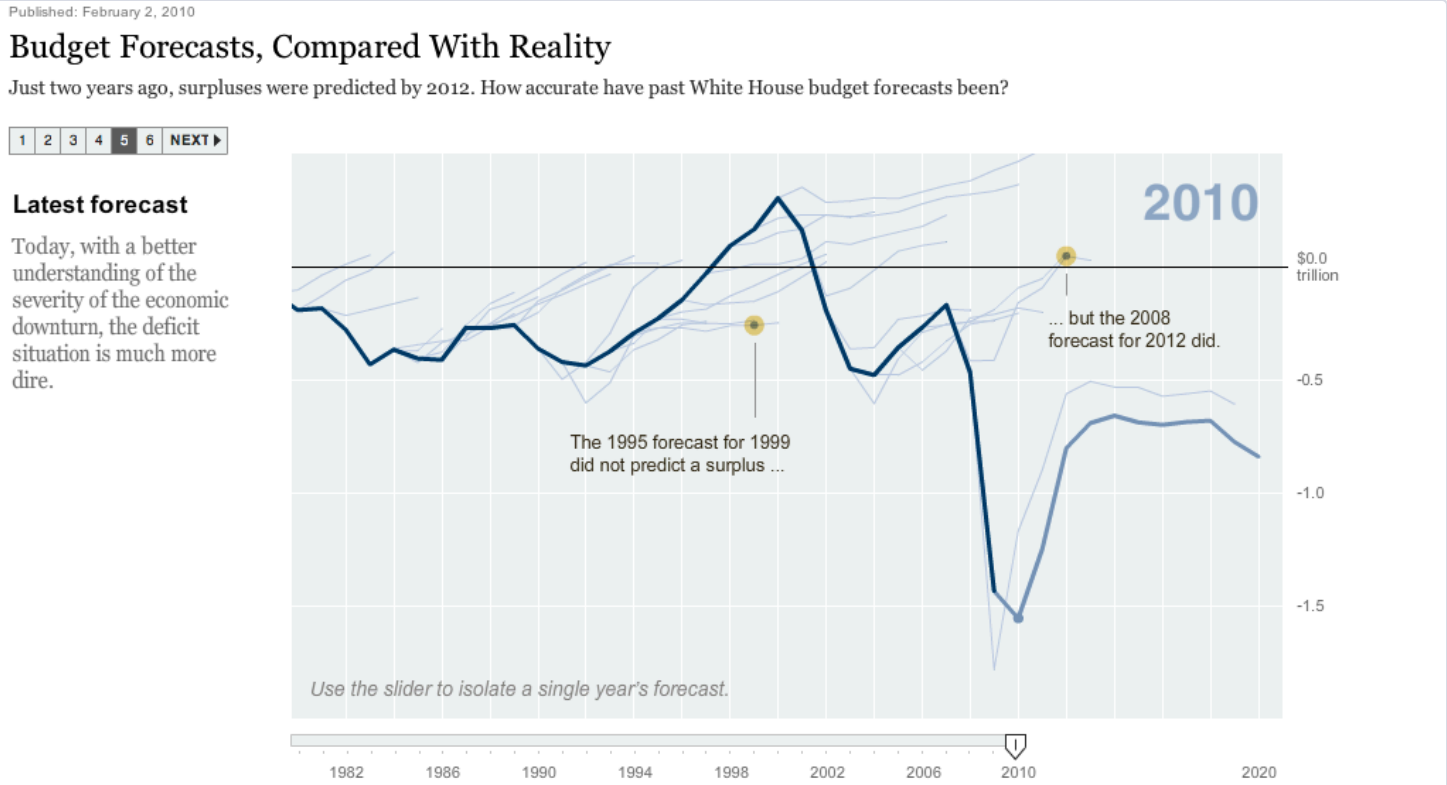
The dark, heavier **actual-results line** stands out from the pale forecasts.

Yellow markers draw attention to two comparison points.

The other forecasts remain visible as context.

Annotations explain the highlighted comparisons

An **annotation** is text or a visual cue attached to a specific part of a visualization to identify it, explain it, or add context.



The New York Times · 2010 · Figure 2 in Segel & Heer · Open archive ↗

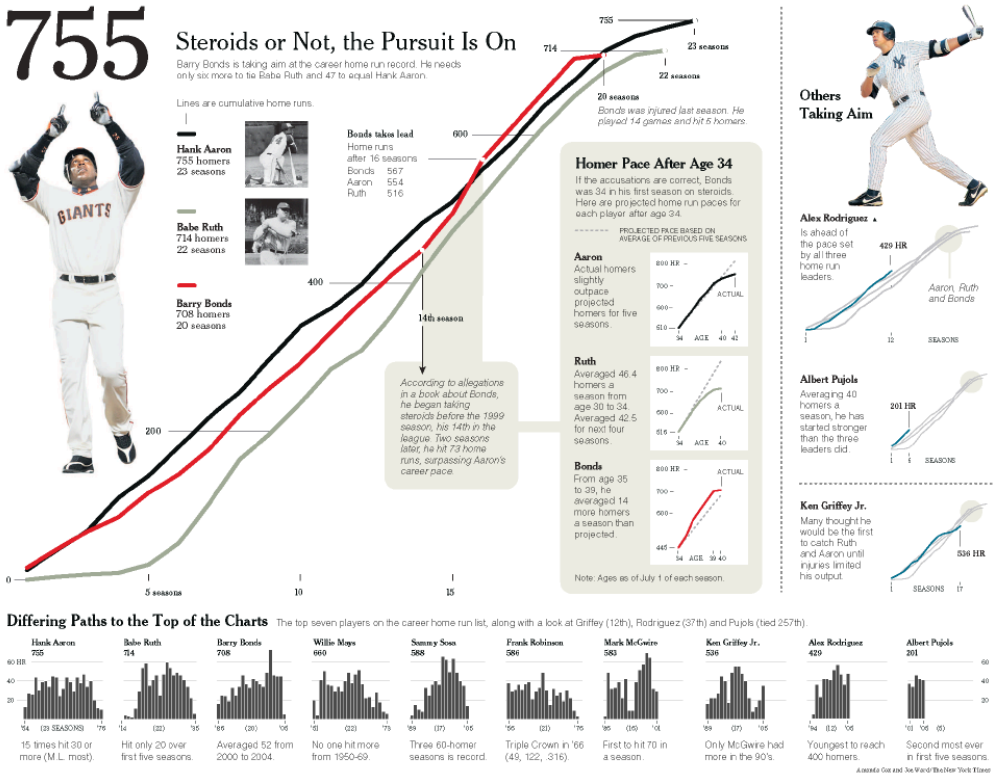
The **1995** and **2008** callouts describe how forecasts differed from later outcomes.

Lines and markers attach each explanation to its location in the chart.

Emphasis attracts attention; annotation gives the comparison meaning.

Layout suggests a reading path

A **reading path** is a suggested order for inspecting a graphic. Visual hierarchy, grouping, alignment, and connectors can guide readers through one static composition.



The New York Times · 2006 · Figure 1 in Segel & Heer · Read the case analysis ↗

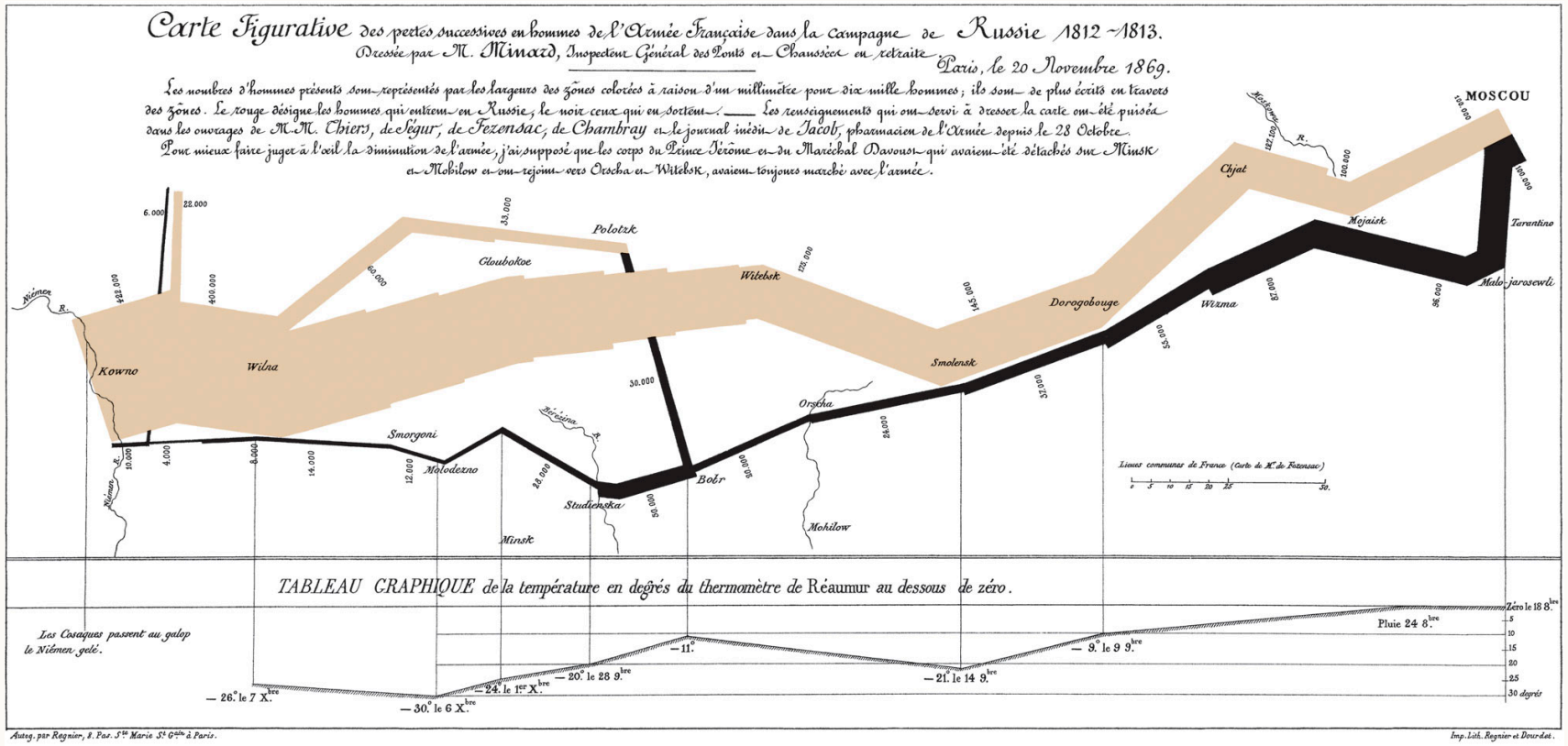
Enter: the large “755” and player figure.

Compare: the main career chart, then the inset linked by a shaded connector.

Extend: the smaller side and bottom panels.

The layout suggests a route; readers can choose another.

One static graphic can carry a narrative



Charles Joseph Minard · 1869 · Public domain · View original ↗

Emphasis
Narrowing bands reveal loss.

Annotation
Place names, counts, dates.

Reading path
Out → back → temperature.

PART 04

Balancing Explanation and Exploration

Use interaction to give readers room to explore within an authored explanation.

Two forces in data stories

- **Author-driven:** The author directs the sequence of views and the explanation.
- **Reader-driven:** The reader chooses which views to explore and in what order.

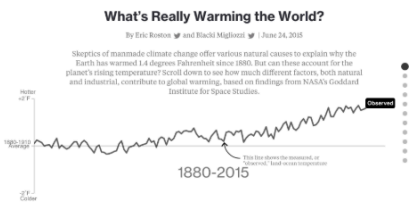
Author-driven (100%)	Reader-driven (100%)
Linear sequence	No prescribed order
Heavy messaging (e.g., annotations)	Minimal messaging
No interactivity	Free interactivity

- **In-between:** A balance between the author-driven and reader-driven, providing limited interactivity within a prescribed narrative. (It's the mainstream of data storytelling nowadays, using limited interactivity within a guided flow.)



video

Narrative visualizations lies between these two sides





dashboard

Three common narrative structures

A **narrative structure** organizes the sequence of views and the opportunities for reader exploration.

Structure	Narrative flow	Interactivity	Example
Martini glass (More author-driven)	Linear sequence → open exploration	After the guided sequence	Gapminder Tools Play, then explore
Interactive slideshow (More in-between)	Stepwise sequence	Within individual scenes	The Yield Curve The New York Times
Drill-down story (More reader-driven)	Non-linear paths from an overview	Reader chooses cases and order	COVID-19 Vaccine Tracker Bloomberg

Structures: Segel & Heer (2010), §4.4 · Read the paper ↗

Martini glass: a guided opening, then exploration

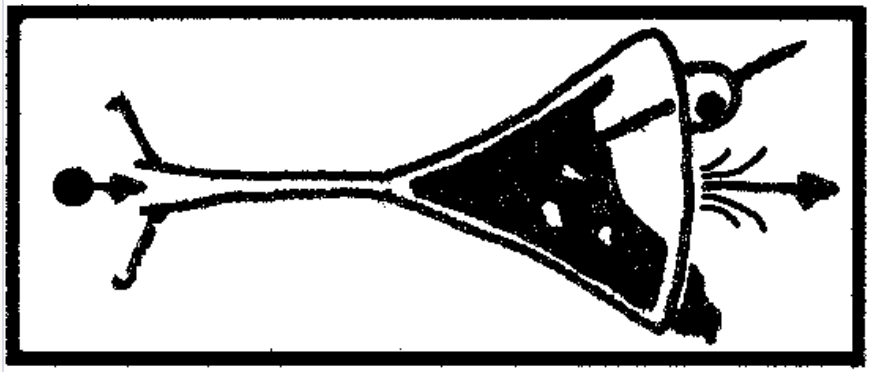
A **martini-glass structure** begins with an author-guided sequence, then opens into reader-driven exploration.

Its name comes from the shape of a martini glass:

Narrow stem: readers follow the author's introduction, observations, and questions in a prescribed order.

Wide mouth: readers enter an open area and independently choose views and comparisons.

More author-driven overall: the author establishes the story and frames readers' later exploration.



Segel & Heer · 2010 · §4.4.1 · Read the paper ↗

Gapminder: play the timeline, then explore



Gapminder Tools · 2025 view · Open tool ↗

1. Play: the stem

First, watch a prepared view play through the years in a fixed order.

2. Explore: the mouth

After playback, choose countries, indicators, and years for your own comparisons.

A shared sequence opens into individual choices.

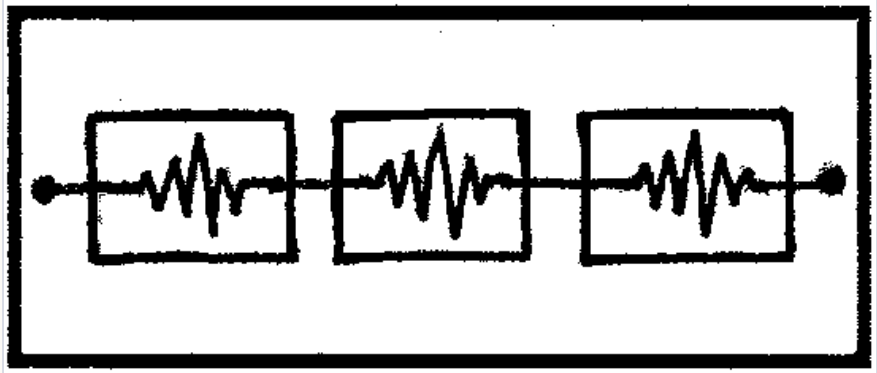
Interactive slideshows embed exploration in scenes

An **interactive slideshow** has an authored order of steps, with data interaction available inside individual steps.

Author guidance: the author prescribes the sequence of scenes and the explanation in each.

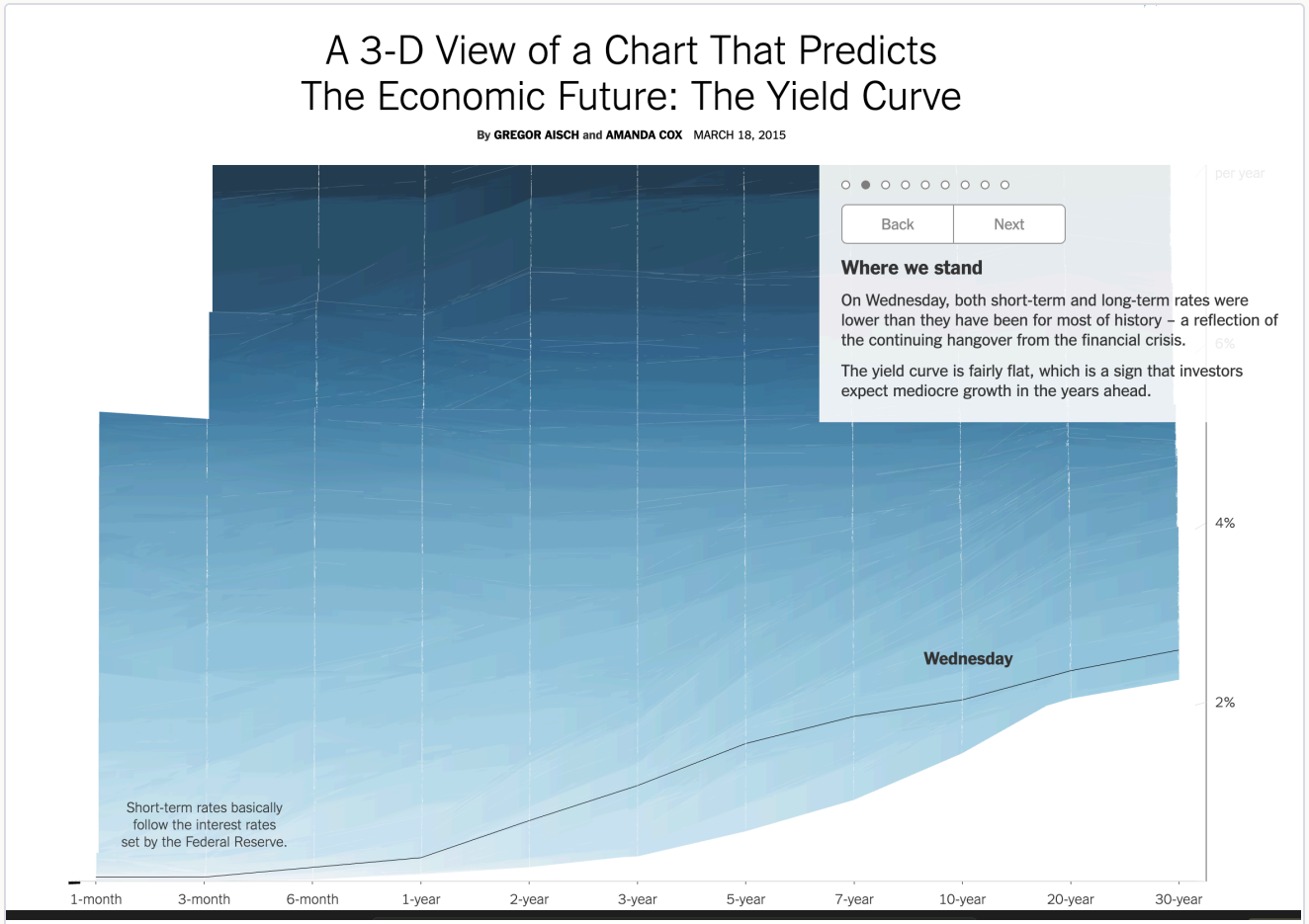
Reader exploration: within a scene, readers can inspect details or compare data before continuing.

More in-between: guidance and exploration alternate within the narrative. Readers make choices inside each scene while the author retains the overall story order.



Segel & Heer · 2010 · §4.4.2 · Read the paper ↗

The yield curve: guided steps, local exploration



Gregor Aisch & Amanda Cox / The New York Times (2015)

Open original ↗ · Author demonstration ↗

The NYT yield-curve story shows how **interest rates** vary by **date** and **borrowing maturity**.

1. Read the step: a guided scene explains how short- and long-term rates compare at one date.

2. Inspect the view: readers rotate the same 3-D chart to inspect the relationship from another angle.

3. Continue the tour: Next moves to another authored viewpoint and explanation, continuing the account across time.

More in-between: the author sets the explanatory sequence. Readers explore within the current scene before continuing.

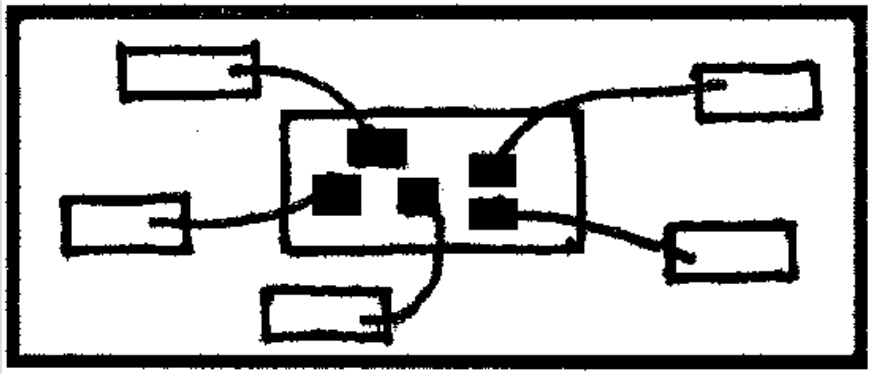
Drill-down stories let readers choose a case

A **drill-down story** introduces a general theme, then lets the reader choose particular cases to reveal more detailed accounts.

Author guidance: the author sets the shared theme and prepares the explanations in each branch.

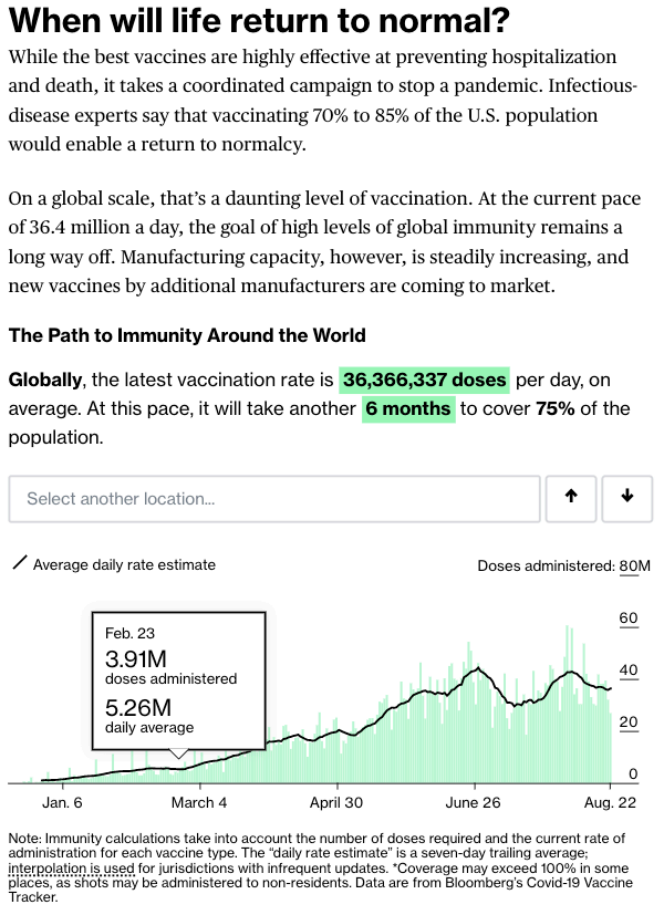
Reader exploration: readers decide which stories to open, when to open them, and in what order.

More reader-driven overall: readers determine their route through the story. There is no single prescribed sequence through all the cases.



Segel & Heer · 2010 · §4.4.3 · Read the paper ↗

Bloomberg: readers choose countries and order



Bloomberg COVID-19 Vaccine Tracker
Historical interface · Open archive ↗

From a global overview to country views

Choose the country: use "Select another location…" to inspect a country's vaccination history.

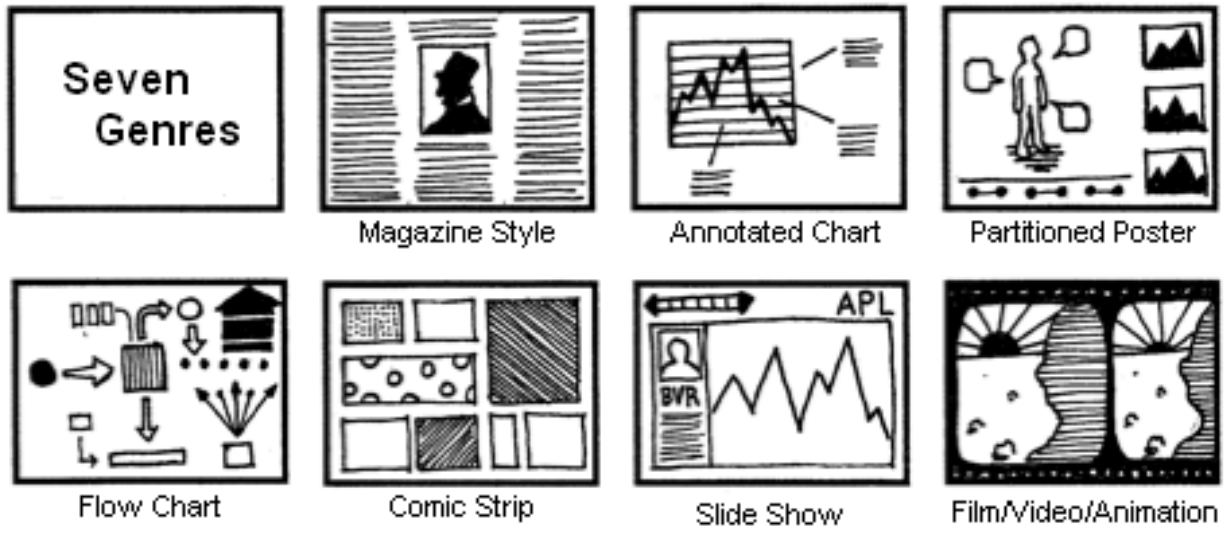
Choose the order: move to another country whenever you want, without following a prescribed sequence.

Possible reading paths:
India → Brazil, or Brazil → India.

More reader-driven: the author supplies shared measures and context; readers decide which country accounts to visit and in what order.

Visual forms of data stories

A **visual genre** is the visual form used to present a data story. Segel and Heer identify seven basic genres.



Edward Segel & Jeffrey Heer · 2010 · Figure 8
Narrative Visualization: Telling Stories with Data ↗

- Magazine style
- Annotated chart
- Partitioned poster
- Flow chart
- Comic strip
- Slide show
- Film / video / animation

These genres are **not mutually exclusive**. A visualization can combine several genres.

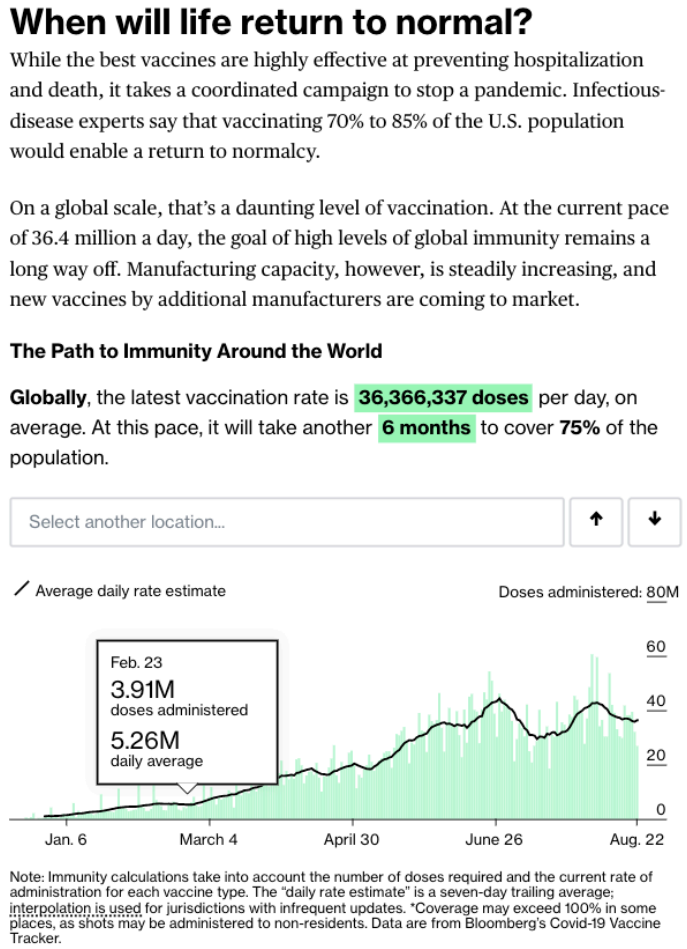
How genres and structures relate

Genres define **appearance**. Structures organize the reader's **experience**.

Structure / control pattern	Possible visual forms	Interaction timing
Martini glass	Annotated chart / Slide show	After the narrative
Interactive slideshow	Slide show / Partitioned poster / Magazine style	Within each scene
Drill-down story	Map / Partitioned poster / Flow chart / Magazine style	User-selected
Pure author-driven	Video / Comic strip	None
Pure reader-driven	Dashboard	User-selected

Concepts: Segel & Heer (2010), §§4.3–4.4 · Read the paper ↗
Pairings are examples, not fixed rules. Interaction timing refers to data exploration.

Genre example: Magazine style



When will life return to normal?

While the best vaccines are highly effective at preventing hospitalization and death, it takes a coordinated campaign to stop a pandemic. Infectious-disease experts say that vaccinating 70% to 85% of the U.S. population would enable a return to normalcy.

On a global scale, that's a daunting level of vaccination. At the current pace of 36.4 million a day, the goal of high levels of global immunity remains a long way off. Manufacturing capacity, however, is steadily increasing, and new vaccines by additional manufacturers are coming to market.

The Path to Immunity Around the World

Globally, the latest vaccination rate is **36,366,337 doses** per day, on average. At this pace, it will take another **6 months** to cover **75%** of the population.

Select another location...

Average daily rate estimate

Doses administered: 80M

Note: Immunity calculations take into account the number of doses required and the current rate of administration for each vaccine type. The "daily rate estimate" is a seven-day trailing average; interpolation is used for jurisdictions with infrequent updates. *Coverage may exceed 100% in some places, as shots may be administered to non-residents. Data are from Bloomberg's Covid-19 Vaccine Tracker.

Bloomberg Vaccine Tracker · historical screenshot · Open original ↗

Magazine style integrates prose and graphics into the reading flow of an article.

Paragraphs establish the topic and introduce the comparison.

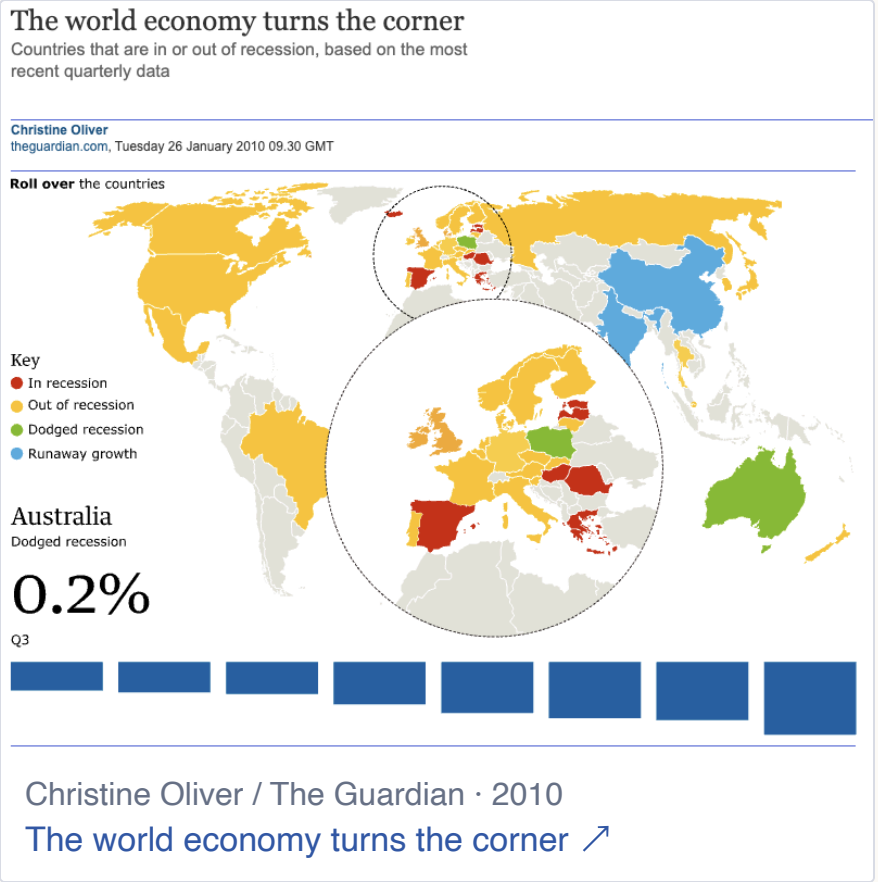
The chart heading connects that discussion to the graphic below it.

The embedded chart and its note develop the explanation in the same column, keeping text and data together.

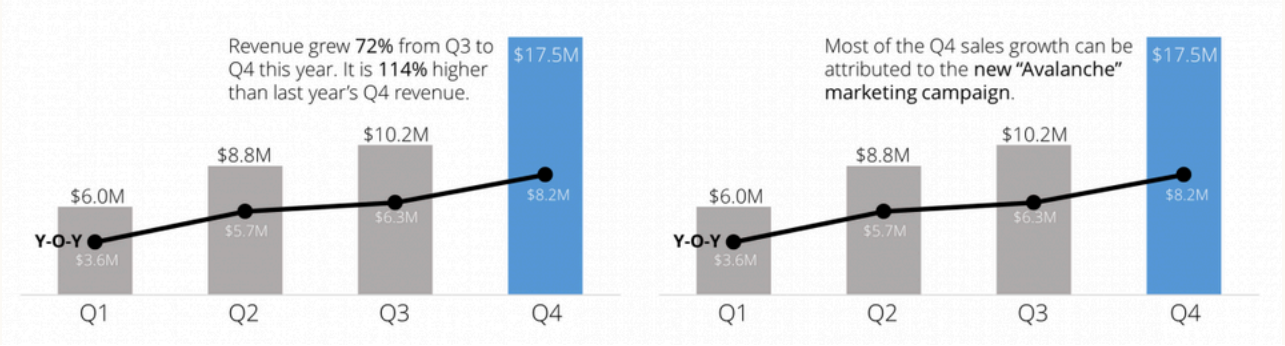
Historical publication state; the numbers and forecasts shown are not current.

Genre example: Annotated chart

An **annotated chart** combines a visualization with text attached to relevant data, helping readers interpret what the chart shows.



Christine Oliver / The Guardian · 2010
The world economy turns the corner ↗

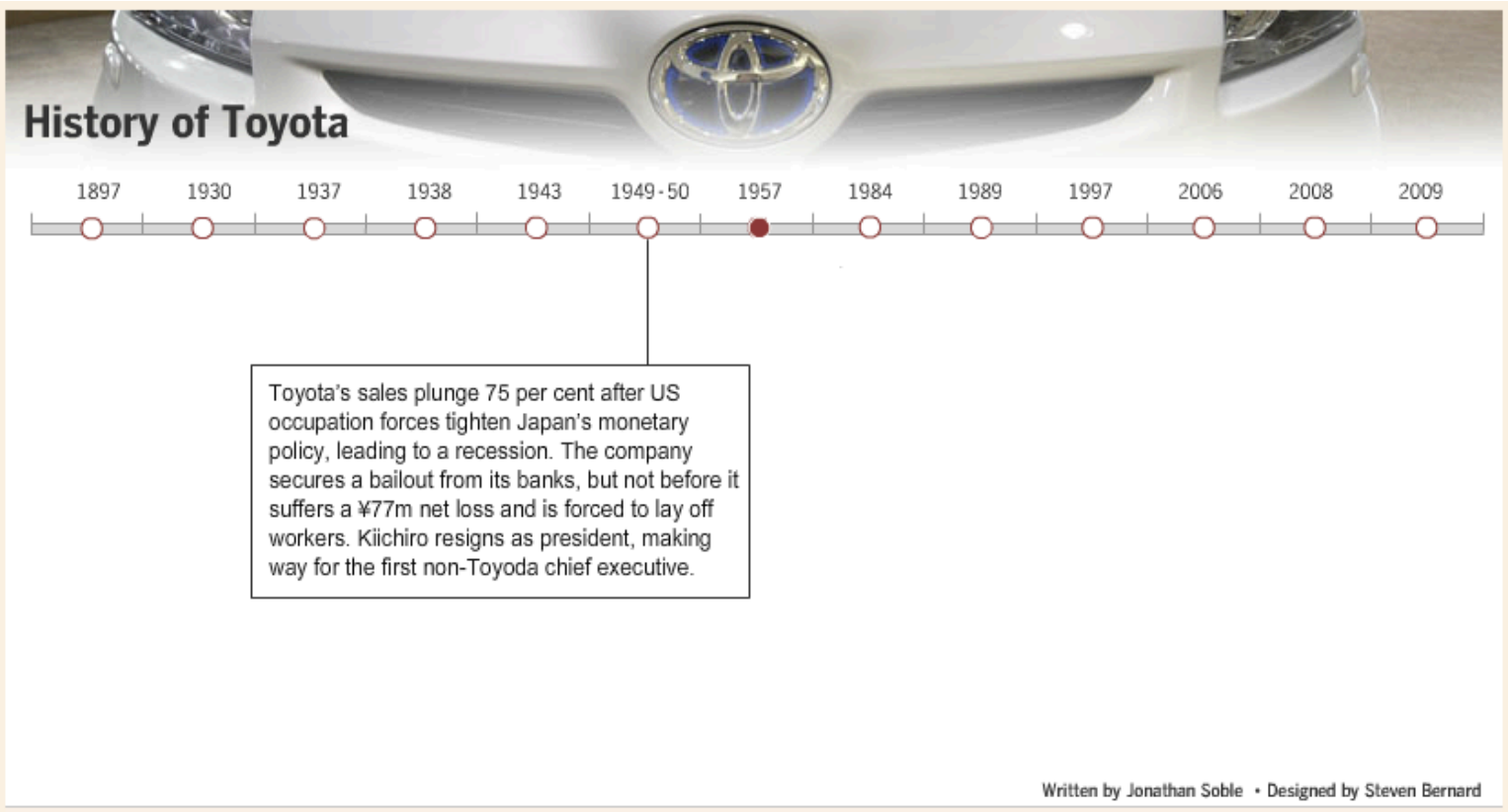


Illustrative revenue comparison · original author uncredited

Left: describe. Restate the increase already visible in the bars.

Right: explain. Add the campaign context to help readers understand the increase.

Genre example: Slideshow



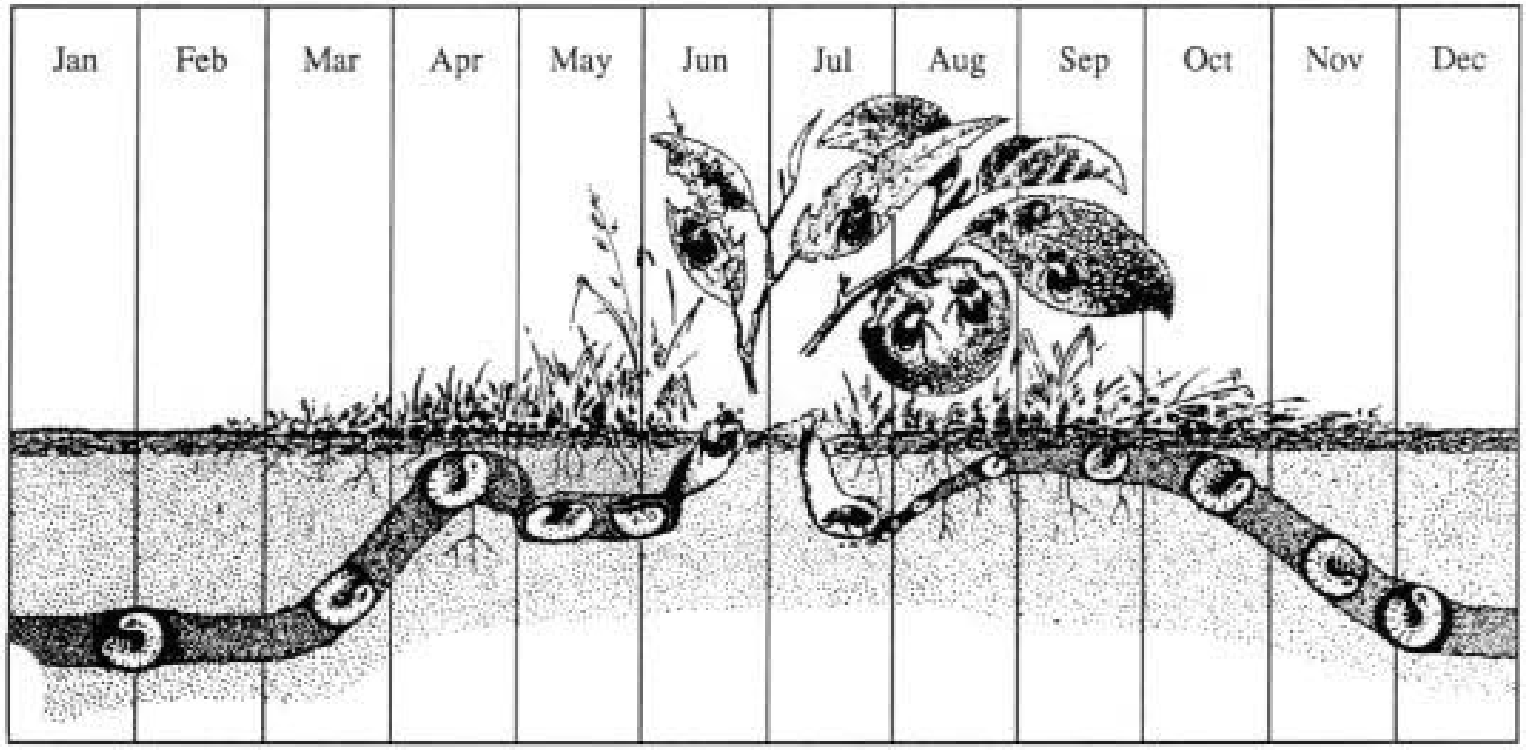
Jonathan Soble · design by Steven Bernard / Financial Times · 2010
Toyota Timeline: A Company History ↗

A **slideshow** presents information through a series of distinct scenes, usually showing one scene at a time.

Here, dated markers organize the history. A text panel explains the selected period.

The timeline keeps each scene within the larger chronology.

Genre example: Comic strip



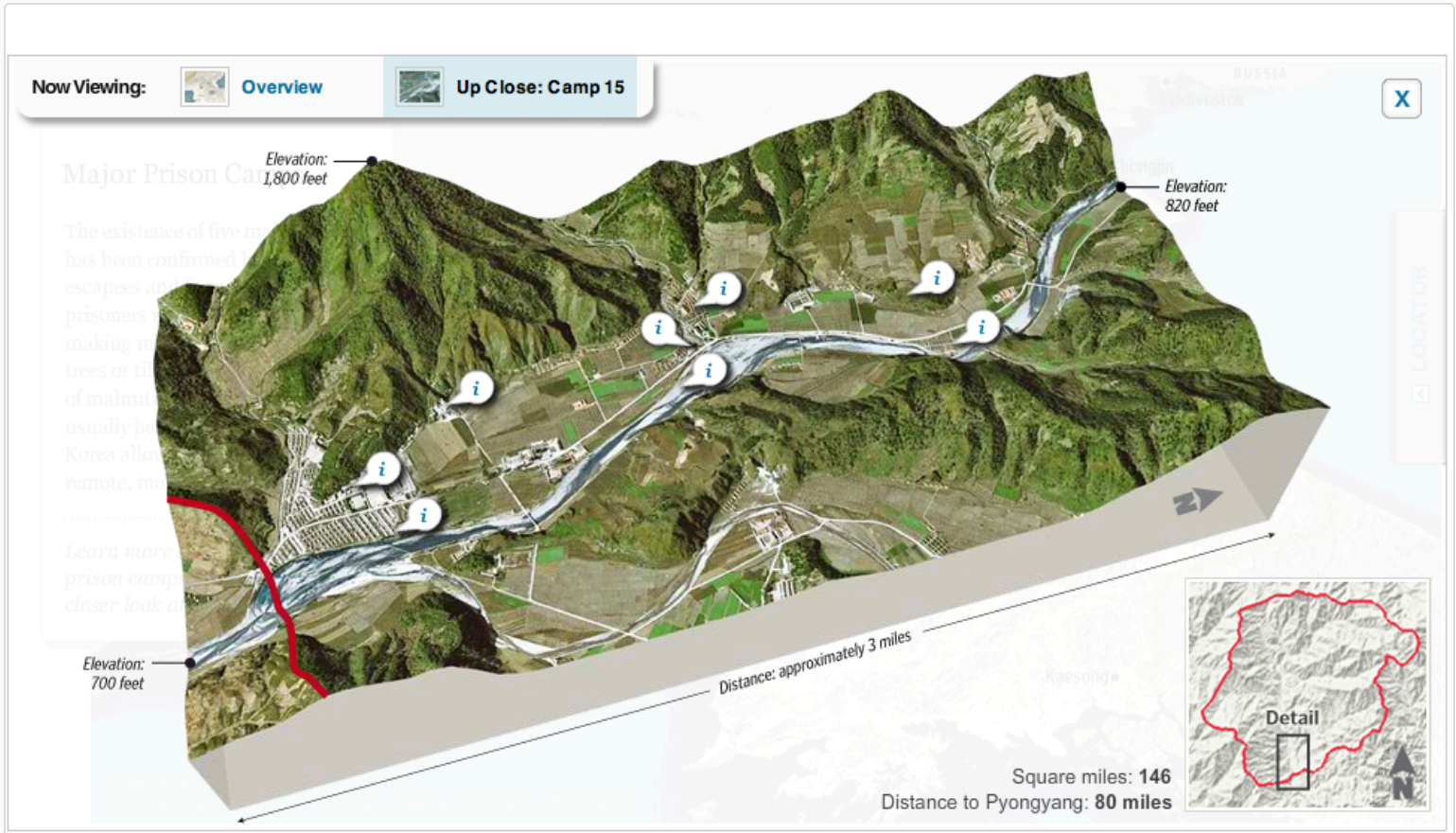
Edward Tufte, The Visual Display of Quantitative Information, 2001
Life Cycle of a Beetle through a Year · Segel & Heer, A43 ↗

A **comic strip** develops a story through adjacent panels. Each panel adds a stage or event to the sequence.

Here, monthly panels show successive stages in the beetle's life cycle.

The ground line connects the panels, while changing forms show development over time.

Genre example: Partitioned poster



K. Downs, B. Harden, L. Heron, L. Karklis & F. Uenuma / The Washington Post · 2009
On the Map: Five Major North Korean Prison Camps ↗

A **partitioned poster** divides one composition into related regions, often combining an overview with detail panels.

The map establishes the setting. Insets and text explain particular places.

Readers relate local details back to the overview within the same composition.

Genre example: Flow chart

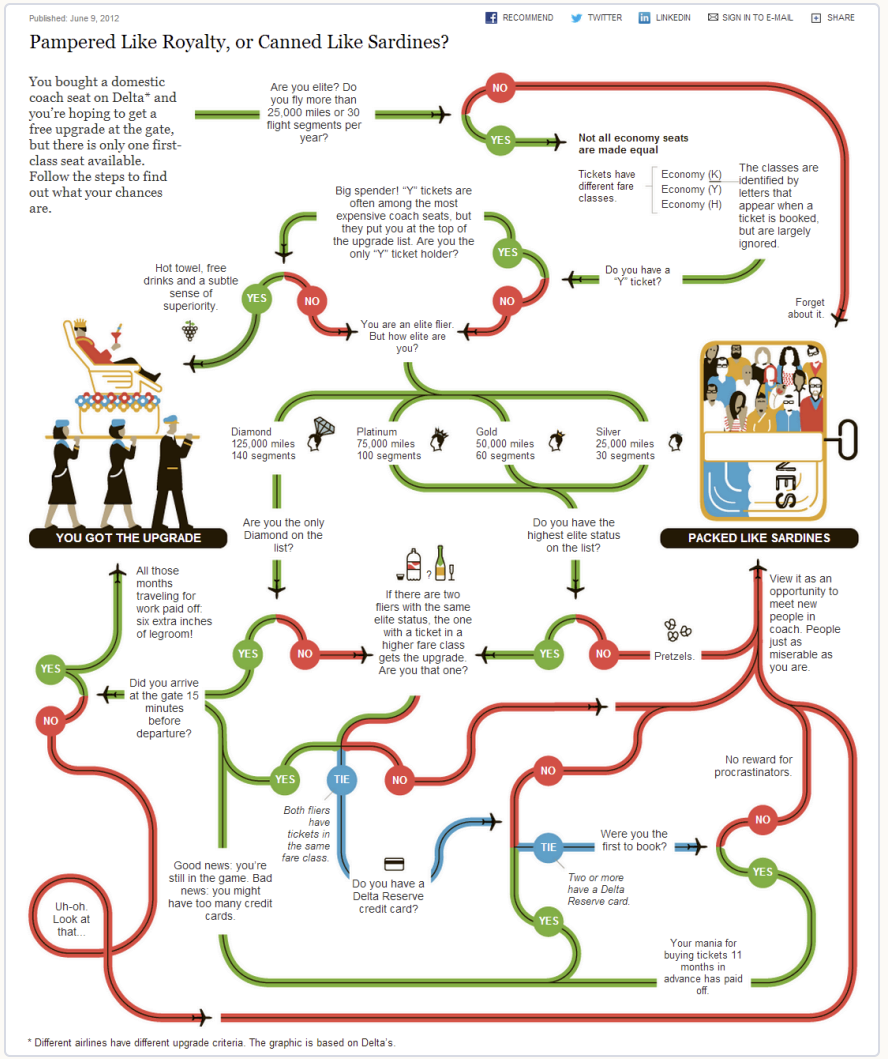
A **flow chart** represents a process or decision path through connected steps, branches, and outcomes.

In this airline-upgrade example, each question directs the reader to another condition or an outcome.

Branches show alternative routes.
Arrows show where to continue.

The author defines the possible paths; readers follow the conditions that apply to them.

The New York Times · June 9, 2012
Pampered Like Royalty, or Canned Like Sardines?
Historical example of airline upgrade rules.



Genre example: Animation



BBC News 中文 · 2022
Open the article and view the animation ↗

Animation presents visual changes over time. In a purely author-driven animation, the author fixes the sequence and transitions.

In this BBC example, the article uses **magazine style**, while an embedded animation presents an authored visual sequence.

One work can combine several genres.

PART 05

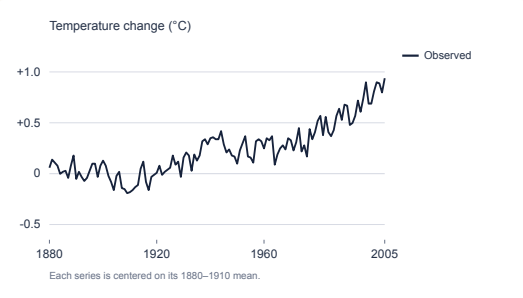
Storyboarding a data story

Bring sequence, visual emphasis, reader choices, and genre together in a plan for each scene.

A storyboard specifies each scene's design

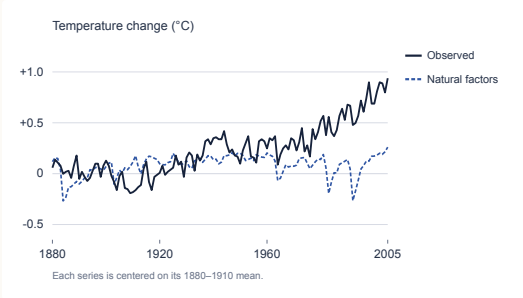
A **storyboard** is a sequence of rough frames that plans what each scene shows, what changes between scenes, and where readers can choose.

1 · CONTEXT



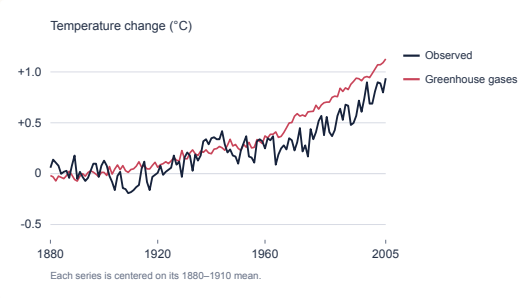
Show the observed line.
Annotate its late-century rise.

2 · QUESTION



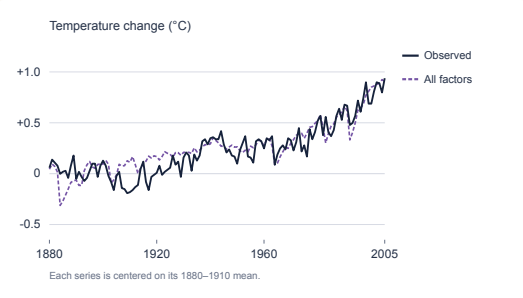
Keep axes and observed line.
Add natural factors and label the gap.

3 · EVIDENCE



Keep the reference.
Replace natural factors with greenhouse gases.

4 · IMPLICATION



Show the combined response.
State the answer, then ask how we respond.

Control: guide readers through frames 1–4, then open factor comparisons.

Storyboard plan using selected Bloomberg stages · reference thumbnails are course replots using NASA GISS data.

Cases and further reading

Cases

Gapminder Tools: play, then explore

Hans Rosling / BBC: 200 years in 4 minutes

Bloomberg: What's Really Warming the World?

R2D3: A Visual Introduction to Machine Learning

NYT: The Yield Curve (2015)

Minard: the Russian campaign graphic (1869)

Bloomberg: COVID-19 Vaccine Tracker

Concepts and discussion

Segel & Heer (2010): Narrative Visualization

Seven genres, author/reader control, and three narrative structures.

Kosara (2016): The scrollytelling scourge

Examples and a critique of scroll-based progression.

Bostock: How To Scroll

Design and implementation considerations for scrolling.